

Technical Communication

Specific objectives

2hrs.

- Introducing the basic concepts of audience, purpose and genre;
- Introducing ethical issues and conducting research

Course contents

- Thinking about audience, purpose and genre;
- Leading and misleading the reader:
 - -ethics at work
 - -ethics for students
 - -how is ethics related to technical communication?
 - -researching technical subjects

Welcome to technical communication class

(A Guide To Technical Communication)

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Why and How of Technical Communication

How do you define Technical Communication?

- This course attempts to answer the question on —
- **How** to do “something” and **Why** to do “it” rather than “**what it is**”

Before the beginning...

Lets begin asking a few questions -

- Do you prefer reading a novel to play Chess? Why?**
- Do you like to play chess or Chinese Checker? Why?**
- How do you prefer to dress up – casual or formal? Why?**
- How do you reply your mom when she scolds you for making your room messy ?**
- Can you write a guideline to your friend on how to make rice pudding? Etc.**

“How(ness)” and “what(ness)”

How do we define Technical Communication?

- **We all begin from asking as “What is ” such & such or so & so, isn’t it?**
- **Technical communication is the process of conveying complex information (technical jargon and complex concepts) in a clear way to a specific audience. It involves the technical content across various mediums such as written documents, presentations, diagrams, videos, and more.**
- **But “What” ends at “is/am/are” in the answer.**
- **This limits our (power of imagination); so lets begin from “how” or “why” .**
- **Or even if we begin from “what” let it be entailed in “why” & “how”.**
- **So, lets begin by asking, “why do we need to study Tech. Comm. at all?”**

Why...?

- **Why do we need to study Tech Comm.?**
- An essential ingredient for the success and quality of any product, service or business.
- If people cannot use something, or cannot find out how to solve problems they might come across, they are less likely to use or buy your product or service.
 - We need to provide more efficient working environment
 - Unfamiliarity with the new machines, devices, programs etc.
 - Consumes a lot of time finding how to do it.
 - It answers six important questions – what, where, when, how, why and who.

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1. Professional Skill Development

It provides practical strategies for developing clear, concise, and effective communication skills, which are essential in professional and technical settings. Mastery of these skills enhances one's ability to convey complex information to diverse audiences.

2. Adaptability in Communication

The emphasizes adapting communication styles to suit different purposes, mediums, and audiences. This is critical in today's dynamic professional environments where technical communicators must tailor their content to meet the needs of specialists and non-specialists alike.

3. Ethical Communication Practices

It highlight the importance of ethical considerations in technical communication. Understanding these principles helps ensure integrity and accountability in creating documents, especially those involving sensitive or impactful information.

4. Multimodal and Technological Awareness

In an increasingly digital world, it equips readers with tools to create multimodal documents, such as reports, proposals, and presentations, using modern technology. This proficiency is indispensable for staying competitive in the workforce.

5. Problem-Solving and Collaboration

It teaches critical thinking and problem-solving skills through the lens of technical communication, encouraging collaboration and teamwork in document design and project management.

6. Relevance Across Disciplines

Technical communication skills are applicable across various fields, including engineering, science, healthcare, business, and education. The course's interdisciplinary approach makes it relevant for professionals and students in numerous domains.

7. Preparation for Real-World Scenarios

The text integrates examples and exercises that mirror real-world technical communication challenges. This hands-on approach prepares readers to tackle tasks such as writing instructions, preparing proposals, and drafting technical reports.

To what action do we call Technical Communication?

- An Activity of preparing & publishing specialized information in a way that allows non-specialists to understand & use the information to accomplish some task.
- Information can be presented
 - Verbally in an oral presentation- a class, over the telephone
 - Often takes a written or visual form
- The main purpose of the technical documents is generally to inform.
- All writing takes place in a specific context, and all writing involves – a writer / speaker, a message, and a reader/listener.

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How does Technical communication differ from other types of writing?

Technical communication is distinguished from other types of writing by its purpose, audience, and approach.

1. Purpose

- **Technical Communication:** Primarily aims to convey specific, factual, and often complex information clearly and efficiently to help the audience perform a task, make a decision, or understand a concept.
- **Other Types of Writing:** Often focus on artistic expression, storytelling, or persuasion, where the goal might be to entertain, inspire, or argue.

2. Audience-Centric Approach

- **Technical Communication:** Tailored to meet the needs of a specific audience, often non-experts or stakeholders with varying levels of familiarity with the subject matter. Audience analysis is critical to ensure clarity and usability.
- **Other Writing:** While audience consideration is essential, it might not be as strictly applied. For example, creative writing often prioritizes the writer's voice or artistic vision over strict audience needs.

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3. Tone and Style

- **Technical Communication:** Focuses on exactness, objectivity, and neutrality. The tone is formal, avoiding ambiguity or emotional appeals unless they serve a practical purpose.
- **Other Writing:** May employ a broader range of tones, from formal to highly personal or emotional, depending on the context (e.g., fiction, essays, or opinion pieces).

4. Format and Structure

- **Technical Communication:** Highly structured, with standardized formats like reports, manuals, memos, and proposals. Documents often include headings, bullet points, tables, and visuals to enhance clarity.
- **Other Writing:** Formats are more flexible, such as essays or novels, where structure is determined by creativity or rhetorical goals.

5. Use of Visuals and Graphics

- **Technical Communication:** Frequently incorporates visuals like charts, diagrams, and infographics to complement textual information and improve understanding.
- **Other Writing:** Visuals are less commonly used or may serve decorative rather than explanatory purposes.

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6. Collaboration

- **Technical Communication:** Often a collaborative process involving teams, as documents may require inputs from multiple experts or departments.
- **Other Writing:** Typically more individualistic, especially in creative or personal genres.

7. Ethical and Legal Considerations

- **Technical Communication:** Strictly matches to ethical and legal standards, ensuring accuracy, transparency, and cooperation with regulations to avoid misinformation or harm.
- **Other Writing:** While ethics are important, there may be more room for interpretation and subjective expression.

8. Dynamic Nature

- **Technical Communication:** context-dependent, often requiring updates as technology or user needs acquire.
- **Other Writing:** May be static, especially in completed works like novels or essays.

So, technical communication is a practical and useful form of writing, distinct from more expressive or rhetorical forms due to its focus on functionality, clarity, and audience engagement in specific contexts.

Introducing the basic concepts of audiences

The concept of audience as an important element in effective technical communication.

1. Audience-Centered Communication

- Technical communication must be tailored to the audience's needs, knowledge, and expectations. Unlike other types of writing that may prioritize the writer's perspective, technical communication places the audience at the forefront.
- **Purpose:** Identify what the audience needs to know or do based on the communication.
- **Clarity:** Use language and formats that the audience can easily understand and use.

2. Types of Audiences

- **Primary Audience:** The immediate readers or users of the document.
- **Secondary Audience:** Individuals who may not directly use the document but need to be informed,

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4. Adapting Content for the Audience

- **Tone:** Use an appropriate tone—formal, neutral, or informal—based on the audience.
- **Style:** Adjust sentence complexity and vocabulary to match the audience's understanding.
- **Structure:** Organize content logically, using headings, lists, and visuals to guide the audience.

5. Anticipating Audience Reactions

- **Biases and Assumptions:** Address potential misunderstandings or preconceived notions.
- **Feedback Mechanisms:** Incorporate opportunities for the audience to ask questions or provide feedback.

6. Multimodal Considerations

- In today's digital age, technical communication often spans multiple formats (e.g., print, digital, audio). It emphasize designing documents that are accessible and usable across different platforms, considering the technological preferences and capabilities of the audience.

By introducing these concepts, understanding and addressing audience needs is not just a component of technical communication—it is its cornerstone. The ability to analyze and adapt to an audience is what distinguishes successful technical communicators.

Technical Comm...

• How do you learn about the users?

know User's needs, expectations, and goals.

You should **include details about the user's** education, lifestyle, interests, values, goals, needs, limitations, desires, attitudes, and patterns of behavior.

- Who they are (profile)- Take an inventory of what you already know about your target user group
- What they do, when and where (context)- the knowledge, experience, and needs of the primary users, or perhaps these users have become more familiar with the product or process over time.
- Why they do it (needs, goals, tasks) ;
- How they do it (experience);
- **Demographic characteristics** – age, sex, race, socio-economic background or class, culture, & first language.
- What they like or dislike (frustrations).

2. Purpose

Every piece of technical communication is driven by a specific purpose. what do you hope to achieve through creating document you are writing? (**Purpose/Aim**) - **to understand** after reading your document:

Informing: Providing accurate and useful information to the audience.

Persuading: Convincing the audience to take specific actions or adopt certain viewpoints.

Instructing: Guiding the audience through processes or tasks.

Identifying the purpose helps writers decide the tone, structure, and content of their communication. For instance, a manual will focus on instructional clarity, whereas a proposal might emphasize persuasion and benefits.

What is genre?

- The word **genre** is a category of communication, shaped by **artistic, musical, or literary composition** characterized by a particular style, form, or content.
- It should be dynamic and situational, evolving with changes in technology and workplace practices.
- Genres also **provide the writer** with **general organizational patterns** that can **help them arrange** what they say and when they say it.
- **For readers**, genres help organize information so that they can more easily **make sense of what they are about** to read.
- Genres are used to **accomplish actions** by selecting one genre over another, **writers convey specific information** that immediately **helps readers to understand the purpose** of the document.
- Examples of technical communication genres include:
 - ☐ Reports: Structured to present findings, analyses, or recommendations.
 - ☐ Proposals: Designed to persuade stakeholders to approve or support a project.
 - ☐ Instructions and manuals: Created to provide step-by-step guidance.
 - ☐ Emails and memos: Focused on efficient and direct communication within an organization.
- Each genre has established conventions in terms of format, tone, and style. Recognizing and adhering to these conventions ensures the communication is professional and easily understood.

Genre and activity sets

- Genre plays in two place i.e. **academic**, and an organization or institution.
- **In education**, the goal of learning is **focused on taking courses, reading books and articles, listening to lectures, working on problems, and participating in discussions- these activities** conclude in **the act of writing**.
- Some of the writing associated with learning is evaluated, self-motivated and understanding
- To know an organization's activities, need to **understand the main genre**
- If puzzled about genre, can ask supervisor, coworkers and avoid weaknesses.
- The following sections provide more information on how to analyze the characteristics of a genre that is new to you.

What are the main genres of technical communication?

Technical writers expect to produce at least some of the following genres:

- Procedures or instructions
- Manuals
- Reports on technical subject matter
- Specifications
- Proposals
- Letters
- Memos
- Oral presentations

Why does genre matter?

- **Genres provide the writer and readers with general organizational patterns** so that they can more **easily make sense** of what they are about to read and write.
- The conventions of genre instructs you to focus on the process of creating text. Literary genre affects the **relationship between writer and reader**.

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- So, there is the interplay between these three elements. For example, understanding the audience informs the choice of language and content, while the purpose shapes the structure and the genre prescribes the format. Effective technical communication requires balancing these elements to craft messages that achieve their intended goals.
- By mastering these core concepts, readers of Technical Communication can improve their ability to create documents that are not only functional but also strategically aligned with the needs of their professional contexts.